

Ruochen Li

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I am a PhD student in Computer Science at Durham University, specializing in deep learning & machine learning, graph representation learning, and spatial-temporal data analysis. My research has focused on leveraging time-series and advanced predictive modeling techniques to solve complex problems in various real-world scenarios, including trajectory prediction for both humans and vehicles, and human motion analysis. I specialize in transforming theoretical concepts into solutions for real-world problems. These research experiences have equipped me with a robust set of skills for extracting actionable insights from complex datasets in various scenarios. My work has resulted in impactful publications in top-tier journals and conferences, underscoring my commitment to advancing research and technology.

KEY SKILLS

Machine Learning & Deep Learning	Spatial-Temporal Data Analysis	Time-series Modeling
Graph Representation Learning	Regression, Clustering & Classification	Pytorch, Tensorflow
Knowledge Integration in Practical Applications	Python, Java, JavaFX, C, C++	MySQL, MongoDB, SQLite
Project Coordination & Team Collaboration	Quantitative & Qualitative Research	Academic Paper Writing
Statistics & Data Science	Large Dataset Exploration	Mathematical Modeling

EDUCATION & PROFESSIONAL DEVELOPMENT

PhD in Computer Science Durham University, UK	2021 – 2025
Thesis: ‘On Deep Learning for Trajectory Prediction using Advancing Representation Learning’ (Pending Submission)	
Relevant Skills: Spatial-temporal modeling, Time-series Modeling, Pedestrian/Vehicle Trajectory Prediction, Human Motion Analysis, Team collaboration, Project management, Academic communication.	
MSc in Advanced Computer Science (Data Analytics) with Distinction University of Leeds, UK	2019 – 2020
Thesis: ‘Predicting Worm Behaviour using Deep Learning Methods’	
Relevant Course: Data Science, Artificial Intelligence, Scientific Computation, Data Mining and Text Analytics, Big Data Systems.	
BSc (Hons) in Computing Science & Artificial Intelligence University of Leeds, UK	2016 – 2019
Dissertation: ‘Smart Video Engagement Analysis’	
Relevant Course: Machine Learning, Algorithms, Linear algebra & probability, Software Engineering, Robotics, Mathematical Concepts, Discrete Mathematics, Distributed Systems.	

PUBLICATIONS

Ruochen Li, Stamos Katsigiannis, Tae-Kyun Kim, Hubert P. H. Shum. “BP-SGCN: Behavioral Pseudo-Label Informed Sparse Graph Convolution Network for Pedestrian and Heterogeneous Trajectory Prediction,” IEEE Transactions on Neural Networks and Learning Systems (TNNLS), IEEE, 2025. Impact Factor: 10.2 Top 10% Journal in Computer Science, Artificial Intelligence	
Ruochen Li, Tanqiu Qiao, Stamos Katsigiannis, Zhanxing Zhu, Hubert P. H. Shum. “Unified Spatial-Temporal Edge-Enhanced Graph Networks for Pedestrian Trajectory Prediction,” IEEE Transactions on Circuits and Systems for Video Technology (TCSVT), IEEE, 2025. Impact Factor: 8.3 Top 10% Journal in Engineering, Electrical & Electronic	
Ruochen Li, Stamos Katsigiannis, Hubert P. H. Shum. “Multiclass-SGCN: Sparse Graph-based Trajectory Prediction with Agent Class Embedding,” IEEE International Conference on Image Processing (ICIP), IEEE, 2022.	
Tanqiu Qiao, Ruochen Li, Frederick WB Li, Hubert P. H. Shum. “From Category to Scenery: An End-to-End Framework for Multi-Person Human-Object Interaction Recognition in Videos,” International Conference on Pattern Recognition (ICPR), Springer, 2024.	
Tanqiu Qiao, Ruochen Li, Frederick WB Li, Yoshiki Kubotani, Shigeo Morishima, Hubert P. H. Shum. “Geometric Visual Fusion Graph Neural Networks for Multi-Person Human-Object Interaction Recognition in Videos,” (Under Review).	

PROFESSIONAL EXPERIENCE

Demonstrator	Oct 2022 – Present
Durham University, UK	
- Postgraduate Module: Text Mining and Language Analytics, Learning from Data, Programming for Data Science. - Undergraduate Module: Data science (Probability, Image Processing and Computer Graphics), Algorithms and Data Structures.	
Student Helper – 21 st ACM SIGGRAPH / Eurographics Symposium on Computer Animation, Durham (Aug – Sep 2022)	
Reviewer – IEE Transactions on Neural Networks & Learning Systems (TNNLS) / IET Computer Vision	